

NOTE: In the original conversation the assistant jumped directly from Volume 07 to Volume 09. Volume 08 ("Discovery & Asset Intelligence") was only proposed as the next milestone, never written out as a full volume. This file therefore contains that proposal and the surrounding conversation so the numbering Volume 01–Volume 56 stays complete.

Volume 08 – Discovery & Asset Intelligence, **where we define exactly how the platform discovers websites, cloud resources, APIs, infrastructure, mobile devices, identities, certificates, and other assets. That discovery capability will become the foundation for Attack Surface Management, vulnerability assessment, SOC, compliance, and every other security function in the platform.**

=====[USER]=====

And will our cyber security operating platform can scan pc/laptop and can find if the pc/laptop is hacked and someone spy on it with undected apps/programs? Good and continue

=====[ASSISTANT]=====

Yes—but with an important engineering clarification.

One of our core principles is:

Project Sentinel will never claim to detect something it cannot technically verify.

That means we will never tell a user *"your computer is hacked"* unless we have evidence supporting that conclusion.

Instead, Project Sentinel will assign a **confidence level** based on collected evidence and explain exactly why.

For example:

Compromise Assessment

This is much more trustworthy than making unsupported claims.

Can Project Sentinel detect spyware or hidden malware?

Yes—but within the capabilities of the operating system and the user's authorization.

For Windows, Linux, and macOS, Project Sentinel can build a very capable host security assessment engine.

Examples include:

Running Processes

Inspect:

- Running processes.
 - Parent/child relationships.
 - Executable paths.
 - Digital signatures (where applicable).
 - Unexpected privilege levels.
 - Resource usage anomalies.
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Persistence Detection

Check common persistence mechanisms such as:

- Startup folders.
- Scheduled tasks.
- Services.
- Login items.
- Systemd services (Linux).
- Registry autoruns (Windows).
- Launch Agents/Daemons (macOS).

These are common locations used by both legitimate software and malware.

Network Activity

Analyze:

- Active network connections.
 - Listening ports.
 - DNS activity (where available).
 - Unexpected outbound communication.
 - Long-lived encrypted sessions.
 - Known malicious IPs/domains (using threat intelligence if enabled).
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File Integrity

Monitor important system areas for:

- New executables.
 - Modified binaries.
 - Changed startup scripts.
 - Unexpected DLL/shared library changes.
 - Suspicious file locations.
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Installed Software Review

Evaluate:

- Installed applications.
 - Unsigned software.
 - Software with known vulnerabilities.
 - Applications requesting excessive privileges.
 - Unknown publishers.
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Security Configuration

Verify:

- Firewall enabled.
 - Disk encryption enabled.
 - Antivirus/endpoint protection status.
 - Secure Boot (where available).
 - OS update status.
 - MFA recommendations for supported accounts.
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User Account Analysis

Review:

- New administrator accounts.
 - Disabled security policies.
 - Unusual group memberships.
 - Weak local account configurations.
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Browser Security

Inspect supported browsers for:

- Risky extensions.
 - Weak password practices (where APIs allow).
 - Insecure settings.
 - Excessive permissions.
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Hidden Malware

This is where we need to be realistic.

Modern malware can use sophisticated techniques such as:

- Rootkits.
- Kernel-mode drivers.
- Bootkits.
- Firmware compromise.
- Memory-only malware.
- Living-off-the-land techniques.

Some of these **cannot be reliably detected from user space alone**.

Therefore, Project Sentinel will never claim:

"We detect every hidden malware."

Instead we define a layered approach.

PROJECT SENTINEL BLUEPRINT

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